

BoustoDoc: A Type-Aware Hybrid Pipeline for Low-Resource Ukrainian Handwritten Document Understanding

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Abstract. Low-resource handwritten document understanding is difficult when a page contains mixed visual content. In RUKOPYS, Ukrainian pages may include handwritten text, printed text, formulas, tables, annotations, images, and graphs. The task requires detecting regions, assigning region types, and transcribing only the regions that need text. We study this setting as a type-aware routing problem rather than as a single optical character recognition (OCR) problem. Our pipeline first uses DocLayout-YOLO to detect regions and predict one of seven region types. It then routes handwritten, printed, and annotation regions to a Cyrillic Transformer-based OCR (TrOCR) branch, routes formula and table regions to a Qwen3-VL branch with low-rank adaptation (LoRA), and assigns empty text to image and graph regions. Under an internal page-level protocol, BoustoDoc achieves an official-style score of 0.8615. With gold boxes and gold region types, hybrid routing reaches a 0.9152 official-style score, outperforming TrOCR-only and Qwen-only routing.

Keywords: Ukrainian documents · Document understanding · Handwritten OCR · low-resource learning · Region routing · Vision-language models

1 Introduction

Handwritten document understanding is more difficult than plain optical character recognition (OCR) because a page often contains many kinds of visual content. A system must not only read text, but also locate regions, assign region types, and decide whether each region should produce a transcription. This is especially important for Ukrainian handwritten documents, where public annotated data and pretrained recognizers are still limited compared with high-resource OCR settings.

We study this problem on RUKOPYS [18], a Ukrainian handwritten document dataset with heterogeneous page sources and seven region types. The dataset contains four sources: National Dictation pages, state-archive documents, university exams/coursework, and school homework. These pages may contain handwritten lines, printed text, formulas, tables, annotations, images, and graphs. The target output is a list of regions, where each region contains a bounding box, a type label, and a text field.

This setting has three main challenges. First, the page layout is not fixed. Some images are clean scans, while others are phone photos of notebooks or homework. Second, the class distribution is imbalanced. Handwritten regions dominate the dataset, while tables, images, and graphs are much rarer. Third, different region types require different recognition behavior. Long handwritten lines, short teacher annotations, mathematical formulas, and tables should not be decoded in the same way.

Therefore, a single OCR model is unlikely to be optimal for every region type. A line recognizer is useful for handwriting and printed text, but it is not designed to produce structured table rows or formula notation. A large vision-language model (VLM) may better follow structured output instructions for formulas and tables, but it can also normalize or infer content if the prompt is not constrained. We address this by using a type-aware hybrid pipeline. The detector predicts both bounding boxes and region types. A rule-based router then sends each region to the branch that matches its type.

Our system, BoustoDoc, has three main components. First, we fine-tune DocLayout-YOLO [22] to detect seven RUKOPYS region types. Second, we adapt a Cyrillic Transformer-based OCR (TrOCR) model [10] for handwritten, printed, and annotation regions using a silver-to-gold curriculum and type-specific decoding settings. Third, we use a Qwen3-VL branch [14] with a low-rank adaptation (LoRA) adapter [5] for formula and table regions, where structured transcription is needed.

The contributions of this work are:

- We formulate RUKOPYS mixed-region document understanding as a type-aware routing problem under low-resource supervision.
- We build an offline hybrid pipeline that combines layout detection, handwritten, printed, and annotation (HPA) OCR, structured formula/table recognition, and schema-valid output assembly.
- We provide controlled internal experiments and ablations showing when type-aware specialization is useful and where the remaining errors occur.

2 Related Work

2.1 Document Layout Analysis

Document layout analysis aims to locate and classify visual regions on a page. Recent datasets and models have improved this task for printed and digital documents. DocLayNet [13] provides a large human-annotated benchmark for document layout analysis. Document Image Transformer (DiT) [9], LayoutLMv3 [6], and Vision Grid Transformer (VGT) [4] show that document-specific visual pre-training improves layout understanding. More recent one-stage detectors such as DocLayout-YOLO [22] and PP-DocLayout [15] provide efficient layout detection for practical document pipelines.

RUKOPYS [18] differs from common printed-layout benchmarks. Its regions are often handwritten line-level boxes, and its pages come from different sources

such as National Dictation pages, state-archive documents, university exams/coursework, and school homework. Rare classes such as tables, images, and graphs are also important because they affect routing and output format. Therefore, we fine-tune DocLayout-YOLO for the seven RUKOPYS classes and use class-aware augmentation to reduce the effect of class imbalance.

2.2 Historical Handwritten Text Recognition

Handwritten text recognition (HTR) has benefited from encoder-decoder models such as TrOCR [10], which combine a visual encoder with a text decoder. These models are effective when the input is already cropped to a text line or region. Historical HTR remains difficult because handwriting style, orthography, image quality, and document format can vary strongly across time and source.

For Cyrillic and related low-resource HTR settings, prior work includes Kazakh offline handwriting recognition [17], Cyrillic OCR [11], and Ukrainian manuscript recognition [16]. These systems mainly focus on text recognition after region boundaries are known. In RUKOPYS, the system must also detect the regions and handle several non-standard content types. Our HPA branch follows this line-level recognition approach, but applies it only to handwritten, printed, and annotation regions.

2.3 Vision-Language Models for Document Understanding

Vision-language models (VLMs) can read visual content and produce structured text outputs. Donut [8] showed that OCR-free document models can solve document understanding tasks without an external OCR engine. Nougat [3] showed that encoder-decoder models can recover structured academic document content such as formulas and tables. More recent models such as mPLUG-DocOwl [19], UReader [20], the Qwen-VL family [1, 14], and PaddleOCR-VL [21] further improve multilingual and structured document understanding.

Full-page VLM parsing can be risky for this benchmark, because exact transcription penalizes normalization, explanation, and hallucinated completion. The model may normalize spelling, add explanations, complete missing text, or solve formulas instead of transcribing them. These behaviors can hurt exact transcription. We use Qwen3-VL [14] only for formula and table crops in the final pipeline. The prompts are source-aware and type-specific, and they explicitly forbid explanation, normalization, and hallucinated completion.

2.4 Curriculum Learning and Parameter-Efficient Adaptation

Curriculum learning [2] trains a model through ordered stages. In OCR, a common strategy is to warm up on larger noisy or synthetic data and then fine-tune on smaller high-quality annotations. This is useful when gold labels are limited but additional pseudo-labeled data is available.

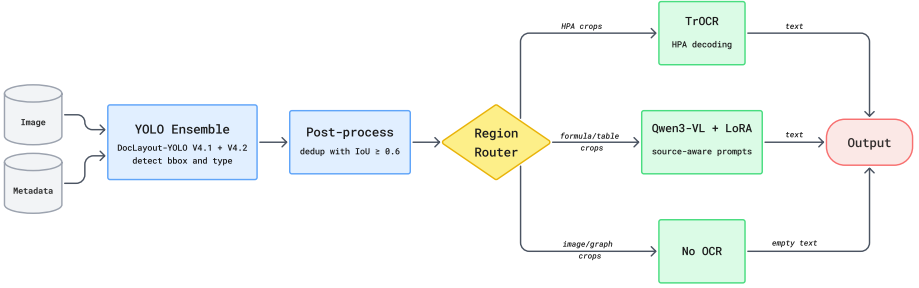


Fig. 1: Inference pipeline overview. Detected regions are post-processed and routed by type. HPA regions go to TrOCR, formula and table regions go to Qwen3-VL, and image or graph regions receive empty text.

Low-rank adaptation (LoRA) [5] is a parameter-efficient fine-tuning method that adapts large models by training low-rank adapter weights instead of updating all model parameters. This is useful for vision-language models, where full fine-tuning is expensive. In our system, the TrOCR branch uses a silver-to-gold curriculum, while the Qwen3-VL branch uses LoRA adaptation. These choices keep the training process practical under limited data and compute.

Overall, prior work provides strong components for layout detection, HTR, and document VLM parsing. In contrast, RUKOPYS requires these components to be combined under exact region-level transcription constraints, where routing by region type becomes important.

3 Methodology

3.1 System Overview

The proposed system receives a document page and outputs a list of regions. Each region contains a bounding box, a type label, and a text field. Figure 1 shows the inference pipeline.

The pipeline follows a type-aware routing design. First, DocLayout-YOLO predicts candidate regions and their seven-way region types. Second, predictions from two detector checkpoints are merged and filtered by non-maximum suppression (NMS). Third, each region is routed according to its predicted type. Handwritten, printed, and annotation regions are sent to the HPA OCR branch. Formula and table regions are sent to the Qwen3-VL branch. Image and graph regions are kept with empty text. Finally, all regions are assembled into a schema-valid comma-separated values (CSV) submission file.

3.2 Region Detection

Region detection is handled by DocLayout-YOLO [22], initialized from a DocStructBench pretrained checkpoint and fine-tuned on the seven RUKOPYS region classes: handwritten, printed, formula, table, annotation, image, and graph.

The detector predicts both the bounding box and the region type, so its output directly defines the routing decision for the later OCR branches.

The detector training data is prepared from gold annotations and rare-class auxiliary samples. Bounding boxes are converted from Common Objects in Context (COCO)-style coordinates to the normalized YOLO format. To reduce the dominance of handwritten regions, we use class-aware augmentation that gives higher priority to rare or structurally important classes such as table, image, graph, and annotation. We use this augmentation mainly for class rebalancing, not just to increase the number of training samples.

At inference time, two detector checkpoints are ensembled. Candidate boxes are generated with a low confidence threshold to preserve recall, then merged with class-agnostic NMS. The remaining regions are clamped to image boundaries, deduplicated, sorted in reading order, and passed to the router.

3.3 HPA OCR Branch

The HPA branch processes handwritten, printed, and annotation regions. It is based on the TrOCR architecture [10] and initialized from a pretrained Cyrillic handwriting checkpoint [7]. Because RUKOPYS contains Ukrainian text, the tokenizer is extended with Ukrainian-specific Cyrillic characters before fine-tuning.

Training follows a silver-to-gold curriculum. The model is first warmed up on auxiliary silver HPA crops, then fine-tuned on gold crops, and finally stabilized with a low-learning-rate gold recovery stage. Checkpoint selection uses a fixed gold validation split and the lowest validation character error rate (CER).

At inference time, each HPA crop is decoded independently. We use type-specific decoding budgets: handwritten regions allow longer outputs, printed regions use a moderate budget, and annotation regions use a short budget to reduce over-generation on small teacher marks or labels.

3.4 Formula and Table OCR Branch

Formula and table regions are routed to Qwen3-VL-8B-Instruct [14] with a LoRA adapter [5]. This branch is used only for structured crop transcription in the final pipeline. Thus, recognition errors in this branch still depend on the detector’s predicted boxes and types.

We route formulas and tables away from TrOCR because their targets are not ordinary text-line transcriptions. Formulas require symbolic structure, indices, operators, and mathematical or chemical notation. Tables require row and cell structure. A line-level HTR decoder may flatten these structures, while a VLM is more suitable for following structured output instructions on these crops.

The Qwen adapter is trained with instruction data that combines crop-level formula/table OCR examples with page-level extraction examples used for auxiliary supervision. During final inference, the model is called only on formula and table crops. The prompts are type-specific and source-aware: formulas are transcribed as visible mathematical or chemical expressions, while tables are returned

as pipe-separated rows. Guardrails are used to discourage explanations, normalization, solved formulas, hallucinated cells, and JSON or Markdown wrappers.

3.5 Post-Processing and Output

After recognition, the system assembles the final region list. Each region contains only three fields: `bbox`, `type`, and `text`. Region types are normalized to the seven valid labels. Image and graph regions are preserved but assigned empty text.

The output for each page is serialized as a JSON list inside one CSV row with two columns: `image` and `regions`. Before writing the final file, the pipeline validates the row count, region keys, bounding-box format, and type labels.

4 Experiments

4.1 Dataset and Evaluation Metrics

Experiments are conducted on the RUKOPYS dataset [18], a Ukrainian handwritten document understanding benchmark with region detection, region-type classification, and text transcription. The dataset contains mixed sources, including National Dictation pages, state-archive documents, university exams/coursework, and school homework. These sources include handwritten text, printed text, formulas, tables, annotations, images, and graphs.

The official test annotations are hidden. Therefore, for controlled evaluation, we split the human-annotated gold training set into internal train, validation, and test splits at the page level. Regions from the same page are never split across different subsets. We use a 70/15/15 split with source-stratified shuffling. We use the internal train split for supervised gold training. We use the validation split for checkpoint selection and hyperparameter tuning. We report final metrics on the internal test split. The silver set is included only as auxiliary noisy training data. It is used for module-specific training stages, such as OCR warm-up and rare-class detector augmentation, but it is not used for checkpoint selection or final metric reporting. All final internal results are computed on the held-out gold internal test split. Table 1 summarizes the resulting data partitions and their role in the experiments.

We report the official composite score used in the Handwritten to Data competition [12] for the RUKOPYS dataset as the main end-to-end metric. It combines type-agnostic detection F1 score (Det-F1) at intersection over union (IoU) ≥ 0.5 , region-type classification accuracy (ClassAcc) on matched regions, region-level character error rate (Region CER), and page-level character error rate (Page CER). The score is computed as:

$$S = 0.15 \text{DetF1} + 0.05 \text{ClassAcc} + 0.30(1 - \text{RegionCER}) + 0.50(1 - \text{PageCER}).$$

Before CER computation, both predictions and ground truth are normalized using the official text-normalization rules. We also report module-level metrics such as detector mean average precision (mAP), crop-level CER, and word error rate (WER) for diagnostic analysis. For completeness, we include the official text-normalization notebook in the supplementary package.

Table 1: Data used for training and evaluation. The gold set is split at the page level for internal evaluation, while the silver set is used only as auxiliary noisy training data.

Data	Split	Pages	Regions	Usage
Gold	Train	931	18,007	Model training
Gold	Validation	199	3,823	Checkpoint selection and tuning
Gold	Internal test	200	3,821	Final internal evaluation
Silver	Auxiliary train	8,207	161,065	Noisy warm-up
Kaggle	Public test	385	Hidden	External leaderboard evaluation

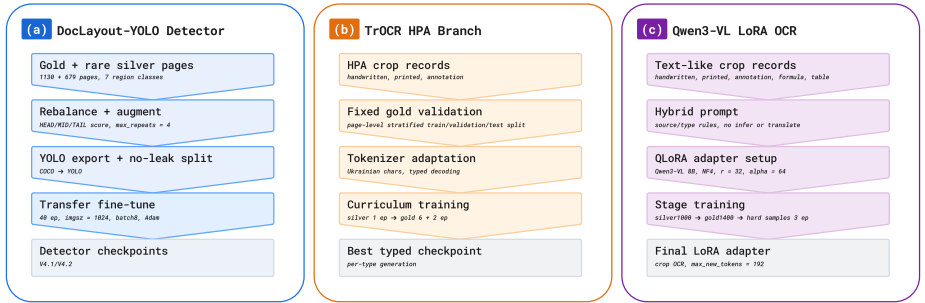


Fig. 2: Training overview for the three modules: DocLayout-YOLO detector fine-tuning, TroOCR HPA curriculum learning, and Qwen3-VL LoRA adaptation for formula/table OCR.

4.2 Implementation Details

Figure 2 summarizes the training workflow for the detector, HPA OCR branch, and Qwen3-VL structured-recognition branch.

The detector is initialized from the DocLayout-YOLO DocStructBench checkpoint and fine-tuned for the seven RUKOPYS region classes. We use batch size 8, image size 1024, Adam optimization, initial learning rate 0.001, weight decay 0.0005, momentum 0.9, two warm-up epochs, and early stopping patience 10. Detector training uses COCO-to-xywh bounding-box conversion, bounding-box validation, and class-aware augmentation. V4.1 uses mosaic 0.1 and stops after 40 epochs, while V4.2 uses mosaic 0.25 and stops after 68 epochs.

The HPA branch handles handwritten, printed, and annotation regions. It is initialized from a Cyrillic HTR checkpoint and fine-tuned with a three-stage curriculum: silver warm-up, gold fine-tuning, and low-learning-rate gold recovery. The learning rates are 3×10^{-5} , 10^{-5} , and 5×10^{-6} . We adapt the tokenizer with Ukrainian-specific characters and use type-specific decoding budgets at validation and inference time.

The Qwen branch handles only formula and table regions after detection and routing. It uses Qwen3-VL-8B-Instruct with LoRA adaptation and 4-bit NormalFloat (NF4) quantized inference. Prompts are type-aware and source-aware:

Table 2: End-to-end gold internal test results under the fixed internal protocol. Higher is better for Det-F1, ClassAcc, and Official Score; lower is better for CER metrics.

System	Det-F1	ClassAcc	Region CER	Page CER	Official Score
Detector only, empty text	0.9221	0.9766	1.0000	0.9628	0.2057
DocLayout-YOLO + TrOCR for all text-like regions	0.9221	0.9766	0.1251	0.1985	0.8504
DocLayout-YOLO + Qwen3-VL for all text-like regions	0.9221	0.9766	0.1347	0.2149	0.8393
DocLayout-YOLO + TrOCR HPA, empty formula/table	0.9221	0.9766	0.2050	0.2771	0.7871
DocLayout-YOLO + Qwen formula/table, empty HPA	0.9221	0.9766	0.9011	0.8439	0.2948
Ours: DocLayout-YOLO + TrOCR + Qwen routing	0.9221	0.9766	0.1061	0.1876	0.8615

All rows use the same detector outputs; differences in Region CER, Page CER, and Official Score come from OCR and routing choices.

formula crops are requested as exact mathematical or chemical expressions, while table crops are requested as pipe-separated rows.

At inference time, the final detector uses a V4.1 + V4.2 checkpoint ensemble. Candidate boxes are generated with confidence 0.001, image size 1024, and at most 300 detections per image. Predictions are merged with class-agnostic non-maximum suppression at IoU 0.55, deduplicated, sorted top-to-bottom and left-to-right, and serialized to `submission.csv`. Each output region contains only `bbox`, `type`, and `text`. Regions of type `image` and `graph` are kept with empty text.

Further implementation details are provided in the supplementary material, including training configurations, prompt templates, inference settings, and output validation checks.

4.3 Main Results

We report results under four evaluation settings. First, the end-to-end internal test evaluates the complete pipeline on the held-out gold pages and is used as the main result. Second, detector-only metrics are computed on the internal detector test split with COCOeval. Third, typed OCR validation evaluates recognizers on gold crops and isolates OCR errors from detector errors. Fourth, the gold-layout routing ablation uses gold boxes and gold region types to isolate the routing policy.

Table 2 reports end-to-end results on the held-out gold internal test split under the fixed internal protocol. These metrics evaluate the full pipeline. They include localization, region typing, region-level transcription, page-level transcription, and the composite score.

Table 3 summarizes module-level diagnostic results. The final detector ensemble reaches 0.7022 mAP@50 and 0.4424 mAP@50:95 under official COCO evaluation (COCOeval) on the 200-page held-out internal detector test split. The HPA OCR branch achieves 0.0954 overall CER on typed validation, with

Table 3: Module-level diagnostic results. Detector results are official COCOeval bounding-box metrics on the held-out internal detector test split; OCR results are crop-level typed validation diagnostics. Higher is better for mAP, average precision (AP), average recall (AR), and Exact Match; lower is better for CER and WER.

Component	Split / Group	Metric	N	Value
DocLayout-YOLO ensemble	detector test	mAP@50	200 / 3,821	0.7022
DocLayout-YOLO ensemble	detector test	mAP@50:95	200 / 3,821	0.4424
DocLayout-YOLO ensemble	detector test	AP@75	200 / 3,821	0.4471
DocLayout-YOLO ensemble	detector test	AR@100	200 / 3,821	0.5877
TrOCR HPA	overall	CER	2,222	0.0954
TrOCR HPA	overall	WER	2,222	0.2913
TrOCR HPA	overall	Exact Match	2,222	0.4043
TrOCR HPA	handwritten	CER	2,147	0.0899
TrOCR HPA	printed	CER	28	0.2121
TrOCR HPA	annotation	CER	47	0.2637
Qwen3-VL	formula	CER	449	0.2019
Qwen3-VL	table	CER	12	0.2359

0.0899 CER on handwritten regions. The printed and annotation subsets are much smaller, so their numbers should be interpreted with caution. In particular, annotation remains the most difficult HPA type because it often contains short, irregular teacher marks and labels.

4.4 Per-Type and Per-Source Analysis

Because RUKOPYS is a mixed-region and multi-source benchmark, aggregate results can hide failure modes. We therefore report performance by region type in Table 4 and by document source in Table 5. The per-type analysis is diagnostic: it shows how errors vary across common HPA regions and rare structured or non-text regions. The per-source analysis shows how performance changes across dictation, archive, university, and school documents.

The rare-class results in Table 4 should be interpreted cautiously. The internal test split contains only 13 table regions, 12 image regions, and 7 graph regions. Therefore, the corresponding AP, Det-F1, and ClassAcc values are useful mainly for diagnosing failure modes, not for making strong claims about robust rare-class performance.

4.5 Ablation Study

We use ablations to isolate the role of each major component: detector ensembling, HPA OCR training, Qwen prompt design, and final type-aware routing. All comparisons are measured under fixed evaluation settings. The routing ablation is treated as an important diagnostic test for the proposed hybrid design.

Detector Ablations Table 6 compares the two detector checkpoints and their ensemble. V4.2 is stronger than V4.1 as a single model, while the V4.1+V4.2 ensemble gives the best AP, Det-F1, and detector-only official score. This suggests that the two checkpoints retain complementary detections.

Table 4: Per-region-type results for the full system. AP values are detector diagnostics from the final ensemble, while counts, Det-F1, ClassAcc, and CER are computed from the final submission under the official matching protocol. Higher is better for AP, Det-F1, and ClassAcc; lower is better for CER.

Type	GT	Pred	AP@50	AP@50:95	Det-F1	ClassAcc	CER	Main failure mode
handwritten	3,149	3,240	0.9415	0.5977	0.9408	0.9873	0.0852	Dense-line duplicate boxes; formula confusion
printed	68	48	0.6069	0.3599	0.7438	0.8889	0.0345	Printed → handwritten confusion
formula	495	557	0.8417	0.5467	0.8953	0.9528	0.2397	Formula/handwritten confusion
table	13	36	0.7424	0.5959	0.5333	1.0000	0.1786	Rare class with many proposals
annotation	77	64	0.4541	0.2446	0.6839	0.6792	0.2630	Missed small annotations; false positives
image	12	17	0.4771	0.4196	0.4138	0.8333	–	Rare class with many proposals
graph	7	8	0.8515	0.3324	0.8750	0.8571	–	Rare class with many proposals

GT denotes ground-truth regions; Pred denotes predicted regions.

Table 5: Per-source internal test results for the full system. RUKOPYS contains noticeable source variation across dictation, archive, university, and school documents.

Source	Pages	Det-F1	ClassAcc	Region CER	Page CER
dictation	54	0.9643	0.9952	0.0567	0.0520
archive	19	0.9511	0.9815	0.1139	0.1539
university	25	0.8904	0.9380	0.2108	0.3459
school	102	0.8923	0.9733	0.1110	0.2269

OCR Ablations Table 7 details the ablation results for the HPA OCR branch. The zero-shot baseline yields an overall CER of 0.6645, confirming a large domain gap. Gold-only fine-tuning reduces the overall CER to 0.1314. However, the printed text CER remains high at 0.4690 due to limited training samples. Introducing a two-stage silver-to-gold setup lowers the overall CER to 0.1081 and cuts the printed CER to 0.2174. The full three-phase curriculum with uniform decoding further decreases the overall CER to 0.1005. This progression suggests that silver data can provide useful warm-up supervision for low-resource categories in this setting. Finally, type-specific decoding gives the lowest overall CER of 0.0954 and the highest exact-match rate of 0.4043. The improvement is mainly driven by handwritten regions, which dominate the HPA validation set. Printed and annotation CER do not improve over uniform decoding, so this setting should be viewed as an overall trade-off rather than a per-type improvement. This ablation is evaluated on gold HPA regions, so it isolates the HPA OCR branch from detector errors.

Table 8 studies the Qwen3-VL formula/table branch. Since formulas and tables have structured targets, we evaluate not only CER but also invalid-output

Table 6: Detector ablation on the 200-page held-out internal detector test split. AP values are official COCOeval diagnostics, while Det-F1 and official score are computed by the official matching protocol at confidence threshold 0.20. Higher values are better for all reported metrics.

Detector	mAP@50	mAP@50:95	Det-F1	Official Score
V4.1 only (confidence 0.20)	0.6163	0.3918	0.9153	0.2037
V4.2 only (confidence 0.20)	0.6961	0.4337	0.9214	0.2050
V4.1 + V4.2 ensemble (confidence 0.20)	0.7022	0.4424	0.9221	0.2057

Table 7: HPA OCR ablation on handwritten, printed, and annotation crops. HW denotes handwritten and Ann. denotes annotation. Lower is better for CER and WER; higher is better for Exact Match.

HPA setting	Overall CER	HW CER	Printed CER	Ann. CER	WER	Exact
Base Cyrillic HTR, zero-shot	0.6645	0.6680	0.6694	0.4865	1.0423	0.0089
Gold-only fine-tuning	0.1314	0.1227	0.4690	0.2631	0.3773	0.3032
Silver warm-up only	0.2095	0.2059	0.3459	0.2659	0.4604	0.2584
Silver → gold	0.1081	0.1023	0.2174	0.2963	0.3306	0.3957
Full curriculum, uniform decoding	0.1005	0.0953	0.2095	0.2586	0.3017	0.3982
Full curriculum, type-specific decoding (final pipeline)	0.0954	0.0899	0.2121	0.2637	0.2913	0.4043

rate. Invalid outputs include explanations, JSON/Markdown wrappers, hallucinated completions, solved formulas, or malformed table rows.

This ablation is evaluated on gold formula and table regions, so it isolates the Qwen branch from detector errors. The official score in this table is therefore used only as a diagnostic score under the same parser and normalization rules, not as the final end-to-end system score. The main quantities of interest are formula CER, table CER, and invalid-output rate.

The zero-shot Qwen3-VL baseline performs poorly on structured crops, with 0.4962 formula CER and 0.4213 table CER. In this ablation, LoRA adaptation gives the largest gain. With a generic prompt, it reduces formula CER to 0.2079 and table CER to 0.2763. Type-specific prompting further improves both structured categories, especially tables, where CER drops to 0.2339. Adding source hints gives the best formula CER of 0.1973 and the highest structural accuracy of 0.9566, suggesting that weak document context can help constrain transcription without changing the output schema. Guardrails did not consistently improve CER or invalid-output rate. We keep them as a conservative decoding choice, but the ablation shows that prompt constraints must be tuned carefully. Overall, the source-aware and type-specific prompt performs well, while guardrails introduce a trade-off between stricter formatting and possible omission of uncertain content.

Table 8: Qwen3-VL formula/table ablation. Invalid-output rate measures whether the model follows the required output format. Lower is better for CER and Invalid Rate; higher is better for structural accuracy (Struct. Acc.) and Official Score.

Qwen setting	Formula CER	Table CER	Struct. Acc.	Invalid Rate	Official Score
Base Qwen3-VL, zero-shot generic prompt	0.4962	0.4213	0.9631	0.0369	0.2850
LoRA, generic prompt	0.2079	0.2763	0.9501	0.0499	0.3120
LoRA, type-specific prompt	0.1996	0.2339	0.9544	0.0456	0.3134
LoRA, source-aware + type-specific prompt	0.1973	0.2383	0.9566	0.0434	0.3137
LoRA, source-aware prompt + guardrails	0.2040	0.2770	0.9501	0.0499	0.3127
LoRA, source-aware + type-specific prompt + guardrails	0.2016	0.2359	0.9523	0.0477	0.3131

Table 9: Routing ablation for the final hybrid pipeline. Higher is better for Det-F1, ClassAcc, and Official Score; lower is better for CER metrics.

Routing strategy	Region CER	Page CER	Official Score
TrOCR for all text-like regions	0.1296	0.1324	0.8949
Qwen3-VL for all text-like regions	0.1382	0.1362	0.8904
TrOCR for HPA, empty formula/table	0.2079	0.2308	0.8222
Qwen for formula/table, empty HPA	0.9011	0.8332	0.3131
Hybrid routing: TrOCR HPA + Qwen formula/table	0.1090	0.1043	0.9152

Routing Ablations Table 9 evaluates the final design choice: using specialized OCR branches by region type. This is the key ablation for the paper because the proposed system is not a single new architecture, but a type-aware hybrid pipeline. This ablation tests whether routing HPA regions to TrOCR and formula/table regions to Qwen3-VL improves recognition when layout annotations are fixed.

For this routing ablation, we use gold bounding boxes and gold region types. This oracle-layout ablation isolates the recognition and routing policy. It complements, but does not replace, the end-to-end comparison in Table 2.

Under this gold-layout ablation, specialization by region type is beneficial. TrOCR applied to all text-like regions reaches a strong score of 0.8949, but it is weaker on structured formula and table content. Qwen3-VL applied to all text-like regions obtains a comparable but slightly lower score of 0.8904, indicating that a VLM can read many crops but is not optimal for ordinary HPA text. Removing formula/table transcription hurts performance, while removing HPA transcription causes a much larger drop because HPA regions dominate the benchmark. Removing formula/table transcription lowers the score to 0.8222. Keeping only Qwen formula/table outputs lowers the score to 0.3131 because HPA regions dominate the benchmark. The hybrid routing strategy achieves the

best result, with 0.1090 region CER, 0.1043 page CER, and 0.9152 official score. This supports the main design choice of routing HPA regions to TrOCR and structured regions to Qwen3-VL instead of forcing a single recognizer to handle all region types.

4.6 Error Analysis

Additional qualitative results are provided in the supplementary material. They include full-page outputs and crop-level examples covering successful cases, near-correct predictions, and representative failures. These examples illustrate the detection-routing-recognition workflow and make common error modes more concrete, including formula-handwriting confusion, annotation-image confusion, malformed table serialization, and routing errors caused by type confusion.

Finally, we analyze representative failure cases. Detector failures are grouped into missed small annotations, duplicate/background proposals, formula/handwritten confusion, and printed/handwritten confusion. Hungarian matching at IoU 0.5 shows that the detector errors are not limited to missed boxes. The ensemble produces 9,043 background false positives in the low-threshold candidate output used for AP-style ranking. Among semantic confusions, formula and handwriting are the most important failure pair, with 20 formula boxes predicted as handwritten and 33 handwritten boxes predicted as formula. Printed-to-handwritten confusion appears 5 times, while handwritten-to-printed confusion is absent in this analysis. Small annotations remain fragile, with 24 missed annotation boxes. We group HPA OCR failures into short annotation errors, Ukrainian character substitutions, punctuation errors, and over-generation. We group formula/table failures into malformed LaTeX, solved or normalized formulas, missing table separators, hallucinated cells, and row-order mistakes.

For the Qwen branch, the main failures are format-related rather than pure character substitutions. Formula errors often occur when the model normalizes notation, simplifies an expression, or produces a solved result instead of copying the visible formula. Table errors are usually caused by missing pipe separators, inconsistent row structure, or hallucinated cells when the crop is sparse or partially occluded. These errors motivate the use of type-specific prompts and output guardrails, but the ablation results also show that prompt constraints must be balanced against the risk of suppressing uncertain visible content. Table 10 summarizes the main failure categories observed across the detector, OCR branches, and full pipeline.

5 Conclusion

We presented a type-aware hybrid pipeline for low-resource Ukrainian handwritten document understanding on RUKOPYS. The system combines DocLayout-YOLO for region detection and type prediction, Cyrillic TrOCR for handwritten, printed, and annotation regions, and Qwen3-VL with LoRA adaptation for formula and table regions. This design treats mixed-region recognition as a routing

Table 10: Qualitative error taxonomy for the final hybrid pipeline.

Module	Error type	Example / Observation
Detector	Missed small annotation	24 missed annotation boxes under Hungarian matching at IoU 0.5.
Detector	Duplicate/background proposals	9,043 background false positives in the low-threshold AP candidate output.
Detector	Formula vs handwritten confusion	20 formula $\rightarrow$ handwritten and 33 handwritten $\rightarrow$ formula confusions.
Detector	Printed vs handwritten confusion	5 printed $\rightarrow$ handwritten and 0 handwritten $\rightarrow$ printed confusions.
TrOCR HPA	Ukrainian character substitution	Visually similar Cyrillic characters and diacritics remain fragile on low-contrast handwriting.
TrOCR HPA	Annotation over-generation	Short marks can be expanded into longer words when the crop contains little textual context.
Qwen3-VL	Formula normalization or solving	The model may rewrite notation or infer a result instead of copying visible symbols exactly.
Qwen3-VL	Malformed table separators	Sparse crops can produce missing pipe separators or inconsistent row lengths.
Full system	Reading-order/page-CER error	Residual duplicate boxes and local ordering mistakes can increase page-level CER.

problem: each detected region is sent to a recognizer that better matches its type, while image and graph regions keep empty text.

Under our internal protocol, the final pipeline reaches an official-style score of 0.8615. In a gold-layout routing ablation, hybrid routing improves over TrOCR-only and Qwen-only recognition for text-like regions. These results provide evidence that type-aware specialization is useful for this dataset. Remaining errors include missed small annotations, formula–handwriting confusion, annotation over-generation, malformed formula outputs, table formatting errors, and page-level ordering issues. Future work should improve rare-class detection, constrained structured decoding, and evaluation on additional Cyrillic document collections.

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